

Geometry summary

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1 Geometry-Aware Curvature Learning with Normalized ϕ_9 Features

We consider a supervised local learning problem for curvature estimation in the level-set representation. Given a local 3×3 stencil of the level-set field, the neural network approximates the dimensionless curvature

$$f_\theta : \mathbb{R}^9 \rightarrow \mathbb{R}, \quad f_\theta(\mathbf{x}_{ij}) \approx h\kappa_{ij}, \quad (1)$$

where h is the mesh size and κ_{ij} denotes the curvature associated with the interface point near the grid node (i, j) .

1.1 Data Generation

In the baseline setting, the training data are generated directly from analytic geometries without solving the reinitialization equation. The geometry family contains circles and ellipses,

$$\mathcal{G} = \{\text{circle}, \text{ellipse}\}. \quad (2)$$

Each geometry is represented by a level-set function ϕ on the uniform Cartesian grid over

$$\Omega = [0, 1]^2, \quad h = \frac{1}{\rho - 1}, \quad (3)$$

where ρ denotes the grid resolution.

For each interface-adjacent grid node (i, j) , we extract the local 3×3 stencil

$$\Phi_{ij} = \begin{bmatrix} \phi_{i-1,j+1} & \phi_{i,j+1} & \phi_{i+1,j+1} \\ \phi_{i-1,j} & \phi_{i,j} & \phi_{i+1,j} \\ \phi_{i-1,j-1} & \phi_{i,j-1} & \phi_{i+1,j-1} \end{bmatrix} \in \mathbb{R}^{3 \times 3}. \quad (4)$$

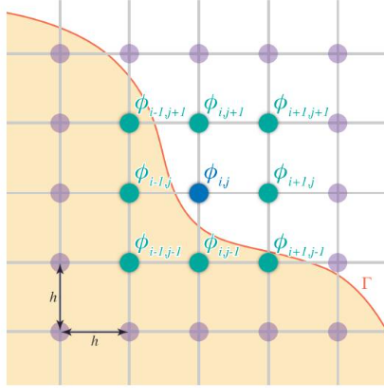


Figure 1: Local 3×3 level-set stencil used for curvature prediction at a grid node (i, j) .

The input feature is obtained by vectorizing the stencil,

$$\mathbf{x}_{ij} = \text{vec}(\Phi_{ij}) \in \mathbb{R}^9, \quad (5)$$

Samples are collected only near the zero level set. Specifically, a node is selected if at least one neighboring grid edge exhibits a sign change:

$$\phi_{i,j}\phi_{i+1,j} \leq 0 \quad \text{or} \quad \phi_{i,j}\phi_{i,j+1} \leq 0. \quad (6)$$

The resulting dataset is

$$\mathcal{D} = \left\{ \left(\mathbf{x}^{(m)}, y^{(m)} \right) \right\}_{m=1}^{N_s}, \quad \mathbf{x}^{(m)} \in \mathbb{R}^9, \quad y^{(m)} \in \mathbb{R}. \quad (7)$$

Both signed-distance and non-signed-distance level-set fields are allowed in the input feature construction, while the target curvature is always computed from the underlying analytic geometry.

1.1.1 Circular Interfaces

For each grid resolution ρ , let $h = (\rho - 1)^{-1}$. Circular interfaces are sampled by

$$\begin{cases} (x - c_x)^2 + (y - c_y)^2 = r^2, \\ r_k = 1.6h + \frac{k}{N_r - 1}(0.5 - 3.6h), & N_r = \left\lfloor \frac{\rho - 8.2}{2} \right\rfloor + 1, & k = 0, \dots, N_r - 1, \\ (c_x, c_y) \sim \mathcal{U}([0.5 - \frac{h}{2}, 0.5 + \frac{h}{2}]^2), & N_v^{\text{circ}} = 12. \end{cases} \quad (8)$$

For each sampled circle, both initial fields

$$\phi_0(x, y) = \begin{cases} \sqrt{(x - c_x)^2 + (y - c_y)^2} - r, & \text{SDF}, \\ (x - c_x)^2 + (y - c_y)^2 - r^2, & \text{non-SDF}, \end{cases} \quad (9)$$

are included. Since the curvature is constant on the interface, the label is simply

$$y = h\kappa = \frac{h}{r}. \quad (10)$$

1.1.2 Numerical Generation of Elliptic-Interface Data

Using the rotated local coordinates

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \cos \psi & \sin \psi \\ -\sin \psi & \cos \psi \end{pmatrix} \begin{pmatrix} x - c_x \\ y - c_y \end{pmatrix}, \quad (11)$$

the interface is

$$\Gamma = \left\{ (u, v) : \frac{u^2}{a^2} + \frac{v^2}{b^2} = 1 \right\}. \quad (12)$$

For each resolution ρ , ellipses are sampled by

$$\begin{cases} a_k = 8h + \frac{k}{N_a - 1}(0.5 - 10h), & N_a = 48, & k = 0, \dots, N_a - 1, \\ \eta = \frac{b}{a_k} \sim \mathcal{U}[0.5, 0.9], & b = \eta a_k, & N_v^{\text{ell}} = 124, \\ (c_x, c_y) \sim \mathcal{U}([0.5 - \frac{h}{2}, 0.5 + \frac{h}{2}]^2), & \psi \sim \mathcal{U}[0, \pi]. \end{cases} \quad (13)$$

Thus the elliptic dataset is sampled uniformly in $(a, b/a)$ rather than in (a, b) . For each sampled ellipse, both

$$\phi_0(\mathbf{x}) = \begin{cases} \phi_{\text{alg}}(u, v) = \frac{u^2}{a^2} + \frac{v^2}{b^2} - 1, \\ \phi_{\text{sdf}}(\mathbf{x}) = \sigma(u, v) \min_{\theta \in [0, 2\pi)} \|(u, v) - (a \cos \theta, b \sin \theta)\|_2, \end{cases} \quad (14)$$

are included, where

$$\sigma(u, v) = \begin{cases} -1, & \frac{u^2}{a^2} + \frac{v^2}{b^2} < 1, \\ +1, & \frac{u^2}{a^2} + \frac{v^2}{b^2} \geq 1. \end{cases} \quad (15)$$

Hence $\phi < 0$ inside Γ and $\phi > 0$ outside. The signed curvature is

$$\kappa_\phi = \nabla \cdot \left(\frac{\nabla \phi}{|\nabla \phi|} \right), \quad (16)$$

which is positive for the convex interfaces considered here.

At a query node $\mathbf{x}_{ij}$ with local coordinates (u_0, v_0) , the label is the dimensionless signed curvature at the closest point $\mathbf{q}(\theta^*) = (a \cos \theta^*, b \sin \theta^*)$, namely

$$\begin{cases} \theta^* = \arg \min_{\theta \in [0, 2\pi)} [(a \cos \theta - u_0)^2 + (b \sin \theta - v_0)^2], \\ g(\theta) = (b^2 - a^2) \sin \theta \cos \theta + a u_0 \sin \theta - b v_0 \cos \theta = 0, \\ y_{ij} = h \kappa_\phi(\theta^*) = h \frac{ab}{(a^2 \sin^2 \theta^* + b^2 \cos^2 \theta^*)^{3/2}}. \end{cases} \quad (17)$$

In practice, (17) is solved by Newton iteration initialized with

$$\theta_0 = \text{atan2}(av_0, bu_0), \quad (18)$$

followed by high-precision refinement; if Newton fails to converge from the analytic seed, a coarse angular sweep is used to reseed the solve (720 points in double precision and 4096 points in the final high-precision fallback).

Finally, for both circles and ellipses, each sample is paired with its sign-flipped counterpart,

$$(\mathbf{f}_{ij}, y_{ij}) \mapsto (-\mathbf{f}_{ij}, -y_{ij}), \quad (19)$$

which applies to both the 9- and 27-dimensional feature representations.

1.2 Dataset

(a) Dataset sizes.

ρ	Circle (blueprints / samples)	Ellipse (blueprints / samples)	Total blueprints	Total samples
256	1,488 / 4,277,752	5,952 / 15,413,576	7,440	19,691,328
266	1,548 / 4,625,460	5,952 / 15,990,988	7,500	20,616,448
276	1,608 / 4,986,540	5,952 / 16,567,956	7,560	21,554,496

(b) Label distribution statistics of $|h\kappa_\phi|$ on the positive half.

ρ	N	σ	P_1	P_{25}	P_{50}	P_{75}	P_{95}	P_{99}	max
256	19,691,328	0.0245	0.0050	0.0089	0.0122	0.0188	0.0425	0.0871	0.6250
266	20,616,448	0.0238	0.0048	0.0086	0.0117	0.0180	0.0409	0.0852	0.6250
276	21,554,496	0.0230	0.0046	0.0082	0.0112	0.0173	0.0394	0.0821	0.6250

Table 1: Dataset sizes and label distribution statistics for the three training resolutions. The sample counts include both initial-field representations and sign-flip augmentation. For the label statistics, N is the total number of signed training samples and σ denotes the standard deviation of the full signed label.

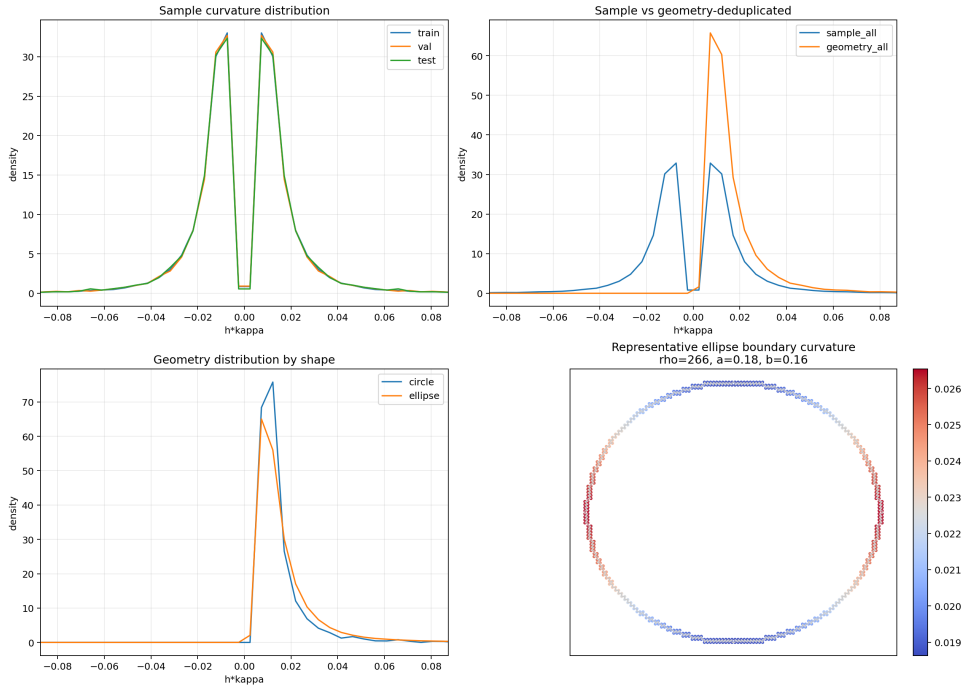


Figure 2: Label distribution for the $\rho = 256$ dataset (19,691,328 samples, 7,440 geometries).

The label distribution is well preserved across the train, validation, and test splits. The sign-flip augmentation makes the sample-level distribution symmetric around zero, with most labels concentrated near small curvature magnitudes and only a small fraction lying in the high-curvature tail. Compared with circles, ellipses introduce a broader curvature range because their curvature varies spatially along the interface.

2 Model Architecture

Table 2: Training hyperparameters.

Group	Parameter	Value
Arch.	Input dim.	9
	Target	$h\kappa$
	Hidden layers	4
	Width	d_h
	Activation	ReLU
	Init.	Xavier
Opt.	Loss	MSE
	Optimiser	Adam
	η	10^{-4}
	(β_1, β_2)	$(0.9, 0.999)$
	ε	10^{-8}
Data	Split	70/15/15
	Normalisation	z-score
Select.	Monitor	val. MAE
	Patience	30 epochs

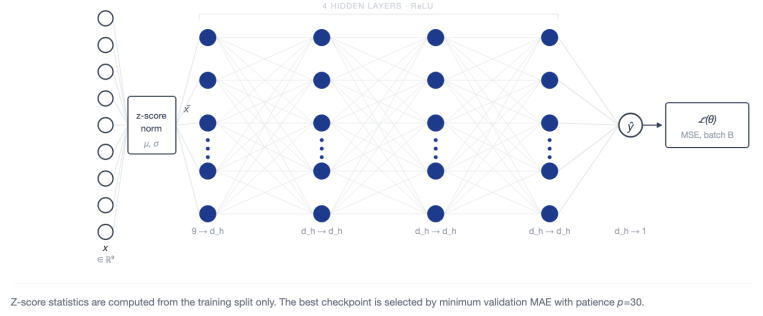


Figure 3: Network architecture.

The predictor is a fully connected MLP mapping the normalised input $\tilde{\mathbf{x}} \in \mathbb{R}^9$ to the dimensionless curvature $h\kappa$. The end-to-end learning pipeline is

$$\mathbf{x} \in \mathbb{R}^9 \xrightarrow{\text{z-score}} \tilde{\mathbf{x}} \xrightarrow{\text{MLP}} \hat{y} = f_{\theta}(\tilde{\mathbf{x}}) \xrightarrow{\text{MSE}} \mathcal{L}(\theta). \quad (20)$$

The network has four hidden layers of uniform width d_h with layer dimensions

$$(d_0, d_1, d_2, d_3, d_4, d_5) = (9, d_h, d_h, d_h, d_h, 1), \quad (21)$$

and computes

$$f_{\theta}(\mathbf{x}) = W_5 \sigma(W_4 \sigma(W_3 \sigma(W_2 \sigma(W_1 \mathbf{x} + b_1) + b_2) + b_3) + b_4) + b_5, \quad (22)$$

where $\sigma(\cdot) = \max(\cdot, 0)$ is the ReLU activation, $W_1 \in \mathbb{R}^{d_h \times 9}$, $W_2, W_3, W_4 \in \mathbb{R}^{d_h \times d_h}$, and $W_5 \in \mathbb{R}^{1 \times d_h}$. All weight matrices are initialised by Xavier uniform and all biases are set to zero:

$$W_{\ell} \sim \mathcal{U}[-a_{\ell}, a_{\ell}], \quad a_{\ell} = \sqrt{\frac{6}{d_{\ell-1} + d_{\ell}}}, \quad b_{\ell} = \mathbf{0}, \quad \ell = 1, \dots, 5. \quad (23)$$

2.1 Model Selection via Early Stopping

At the end of each training epoch t , the validation mean absolute error is evaluated:

$$\text{MAE}_{\text{val}}^{(t)} = \frac{1}{N_{\text{val}}} \sum_{n=1}^{N_{\text{val}}} |f_{\theta_t}(\tilde{\mathbf{x}}_n) - y_n|. \quad (24)$$

If $\text{MAE}_{\text{val}}^{(t)}$ strictly improves over the current best, the model checkpoint is updated and the non-improvement counter is reset to zero; otherwise the counter is incremented. Training terminates once the counter reaches the patience $p = 30$. The final deployed model is the checkpoint attaining the lowest validation MAE:

$$\theta^* = \theta_{t^*}, \quad t^* = \arg \min_{t \geq 1} \text{MAE}_{\text{val}}^{(t)}. \quad (25)$$

This procedure ensures that the reported test performance is attributed to the model selected by out-of-sample accuracy rather than the final training iterate.

2.2 Evaluation Metrics

All metrics are computed on the held-out test set. Let $e_n = f_{\theta^*}(\tilde{\mathbf{x}}_n) - y_n$ denote the point-wise prediction error for sample $n = 1, \dots, N_{\text{te}}$. The reported metrics are

$$\text{MSE} = \frac{1}{N_{\text{te}}} \sum_{n=1}^{N_{\text{te}}} e_n^2, \quad \text{RMSE} = \sqrt{\text{MSE}}, \quad (26)$$

$$\text{MAE} = \frac{1}{N_{\text{te}}} \sum_{n=1}^{N_{\text{te}}} |e_n|, \quad \text{MaxAE} = \max_{1 \leq n \leq N_{\text{te}}} |e_n|. \quad (27)$$

Together, MSE/RMSE characterise average squared error, MAE provides a scale-consistent measure of central tendency, and MaxAE captures worst-case deviation—jointly profiling both the bulk and tail behaviour of the approximation error.

3 Test flower

For the flower test set, the level-set field is defined on a uniform Cartesian grid

$$\Omega = [-L, L]^2, \quad x_i = -L + ih, \quad y_j = -L + jh, \quad h = \frac{2L}{N-1}.$$

The exact flower interface is written in polar form as

$$r(\theta) = b + a \cos(p\theta), \quad \Gamma(\theta) = r(\theta)(\cos \theta, \sin \theta),$$

and the initial field is

$$\phi_{i,j}^0 = \sqrt{x_i^2 + y_j^2} - r(\theta_{i,j}), \quad \theta_{i,j} = \text{atan2}(y_j, x_i).$$

In the current project, the two canonical flower shapes are

$$(a, b, p) = (0.05, 0.15, 3) \quad (\text{smooth}), \quad (a, b, p) = (0.075, 0.15, 3) \quad (\text{acute}).$$

For generic resolution control, one may view the setup as

$$h \approx \frac{1}{\rho_{\text{model}} - 1}, \quad L \approx a + b + 2h, \quad N \approx \text{round}\left(\frac{2L}{h}\right) + 1,$$

while the commonly used $\rho_{\text{model}} \in \{256, 266, 276\}$ cases are stored as fixed repository presets.

Starting from ϕ^0 , the field is reinitialized by

$$\frac{\partial \phi}{\partial \tau} + S(\phi)(|\nabla \phi| - 1) = 0, \quad S(\phi) = \frac{\phi}{\sqrt{\phi^2 + (\varepsilon_s h)^2}}, \quad \varepsilon_s = 2.5.$$

The project default uses a dynamic sign field, namely $S(\phi)$ is recomputed from the current RK stage rather than frozen at ϕ^0 . The Hamiltonian is discretized by a Godunov upwind flux with fifth-order HJ-WENO one-sided derivatives, and pseudo-time marching uses third-order TVD Runge-Kutta with

$$\Delta \tau = \text{CFL} \cdot h, \quad \text{CFL} = 0.5.$$

By default, snapshots are recorded at iterations

$$n \in \{1, 2, \dots, 20\},$$

and $n = 0$ may additionally be included to denote the raw field before reinitialization.

At each recorded iteration n , samples are taken from the current-interface nodes of $\phi^{(n)}$, i.e. grid nodes adjacent to a sign-changing horizontal or vertical edge:

$$(i, j) \in \mathcal{I}(\phi^{(n)}) \iff \phi_{i,j}^{(n)} \phi_{i\pm 1,j}^{(n)} \leq 0 \text{ or } \phi_{i,j}^{(n)} \phi_{i,j\pm 1}^{(n)} \leq 0,$$

with two boundary layers excluded in implementation. For each retained node, the 3×3 stencil is encoded in training order as

$$\mathbf{x}_{i,j} = [\phi_{i-1,j+1}, \phi_{i,j+1}, \phi_{i+1,j+1}, \phi_{i-1,j}, \phi_{i,j}, \phi_{i+1,j}, \phi_{i-1,j-1}, \phi_{i,j-1}, \phi_{i+1,j-1}]^\top \in \mathbb{R}^9.$$

The target is not obtained from a finite-difference curvature formula. Instead, for the current node location

$$\mathbf{z}_{i,j} = (x_i, y_j),$$

the nearest-point projection angle on the exact flower curve is computed by

$$\theta_{i,j}^* = \arg \min_{\theta \in [0, 2\pi)} \|\mathbf{z}_{i,j} - \Gamma(\theta)\|_2^2,$$

and the analytic curvature of the polar curve is

$$\kappa(\theta) = \frac{r(\theta)^2 + 2r'(\theta)^2 - r(\theta)r''(\theta)}{(r(\theta)^2 + r'(\theta)^2)^{3/2}}, \quad r'(\theta) = -ap \sin(p\theta), \quad r''(\theta) = -ap^2 \cos(p\theta).$$

The label is then defined as the dimensionless quantity

$$y_{i,j} = h \kappa(\theta_{i,j}^*).$$

Therefore, the flower test set in this project can be summarized as the mapping

$$\text{current reinitialized } 3 \times 3 \text{ stencil of } \phi \mapsto h\kappa$$

where the ground-truth curvature is obtained from analytic projection onto the exact flower interface. No sign-flip augmentation is applied in this flower test-data pipeline.

4 Experiment 1

4.1 Analysis

4.1.1 Model setting

For each interface-adjacent grid node (i, j) , the local 3×3 stencil is ordered as

$$\phi_9 = (\phi_{i-1,j+1}, \phi_{i,j+1}, \phi_{i+1,j+1}, \phi_{i-1,j}, \phi_{i,j}, \phi_{i+1,j}, \phi_{i-1,j-1}, \phi_{i,j-1}, \phi_{i+1,j-1})^\top \in \mathbb{R}^9. \quad (28)$$

The target variable is the dimensionless curvature $y^{(m)} = h\kappa \in \mathbb{R}$.

1. Raw stencil model. The first model directly uses the nine level-set values as input:

$$\mathbf{x}^{(m)} = \phi_9 \in \mathbb{R}^9, \quad y^{(m)} = h\kappa \in \mathbb{R}. \quad (29)$$

2. Normalized stencil model. The second model scales the stencil by the grid spacing h , giving a dimensionless input:

$$\mathbf{x}^{(m)} = \frac{\phi_9}{h} \in \mathbb{R}^9, \quad y^{(m)} = h\kappa \in \mathbb{R}. \quad (30)$$

3. Normal-augmented stencil model. The third model augments ϕ_9/h with the unit-normal components computed at the nine stencil nodes:

$$\mathbf{x}^{(m)} = \left(\frac{\phi_9}{h}, \mathbf{n}_9^x, \mathbf{n}_9^y \right) \in \mathbb{R}^{27}, \quad y^{(m)} = h\kappa \in \mathbb{R}. \quad (31)$$

Here

$$\mathbf{n}_k = \frac{(\phi_{i+p_k+1,j+q_k} - \phi_{i+p_k-1,j+q_k}, \phi_{i+p_k,j+q_k+1} - \phi_{i+p_k,j+q_k-1})^\top}{\left[(\phi_{i+p_k+1,j+q_k} - \phi_{i+p_k-1,j+q_k})^2 + (\phi_{i+p_k,j+q_k+1} - \phi_{i+p_k,j+q_k-1})^2 \right]^{1/2}}, \quad k = 0, \dots, 8, \quad (32)$$

with $\mathbf{n}_9^x = (n_0^x, \dots, n_8^x)^\top$ and $\mathbf{n}_9^y = (n_0^y, \dots, n_8^y)^\top$. “

4.1.2 Baseline vs phi/h

Figures 4 and 5 compare the raw- ϕ model and the normalized- ϕ/h model on static ellipses with increasing eccentricity. Both models reproduce the overall trend of $h\kappa(\theta)$, while their main errors remain concentrated near the high-curvature endpoints as the ellipse becomes more eccentric.

However, the normalized- ϕ/h input does not yield a more stable prediction in this experiment. Compared with the raw- ϕ model, the ϕ/h model exhibits more oscillatory model-to-analytic errors in several cases, with larger local fluctuations and more pronounced error peaks. This indicates that scaling the stencil values by h does not automatically improve the numerical conditioning of the learned curvature map. Instead, under the same-resolution setting, the enlarged input scale may increase the model sensitivity to local stencil perturbations.

Figures 6 and 7 compare the raw- ϕ model and the normalized- ϕ/h model on smooth and acute flower interfaces at resolution 256 after reinitialization. Both models reproduce the main curvature profile of $h\kappa(\theta)$, and the dominant discrepancies are localized near the high-curvature tip regions. Compared with the raw- ϕ baseline, the normalized- ϕ/h model does not show a clear improvement in error stability or peak-error reduction. The model-to-analytic error retains a similar spatial distribution, with residual peaks concentrated around the same rapidly varying curvature regions. This indicates that ϕ/h normalization alone preserves the global prediction behavior but does not substantially improve robustness for flower geometries, especially in the acute case.

4.1.3 Baseline vs gradient

Figures 8 and 9 compare the baseline stencil model and the gradient-augmented model on static ellipses with increasing eccentricity. The baseline model captures the global curvature profile, but its errors are concentrated near the high-curvature endpoints, where $h\kappa(\theta)$ varies most rapidly.

With gradient augmentation, the model-to-analytic error is reduced over most boundary regions, indicating improved overall generalization across ellipse geometries. However, the error reduction is not pointwise uniform: the largest residuals still occur near the high-curvature endpoints. Thus, gradient features improve the global

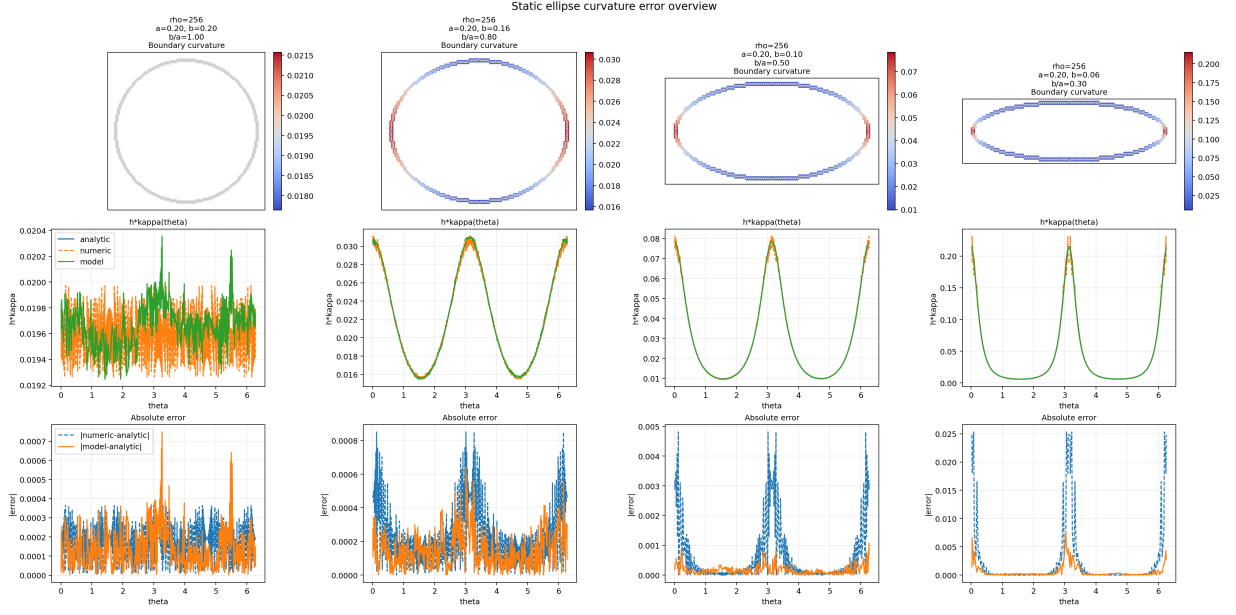


Figure 4: Static ellipse curvature prediction of the raw- ϕ model at resolution 256. The model captures the global curvature profile, while the dominant errors are localized near the high-curvature endpoints.

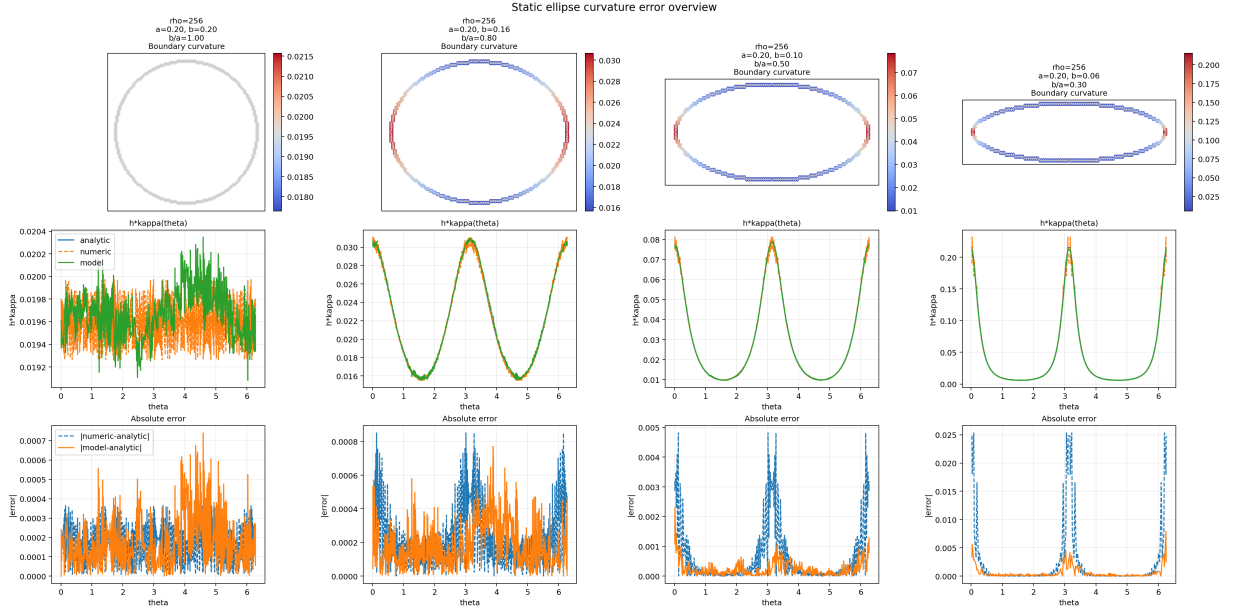


Figure 5: Static ellipse curvature prediction of the normalized- ϕ/h model at resolution 256. The global curvature trend is preserved, but the model-to-analytic error becomes more oscillatory and less stable than the raw- ϕ baseline.

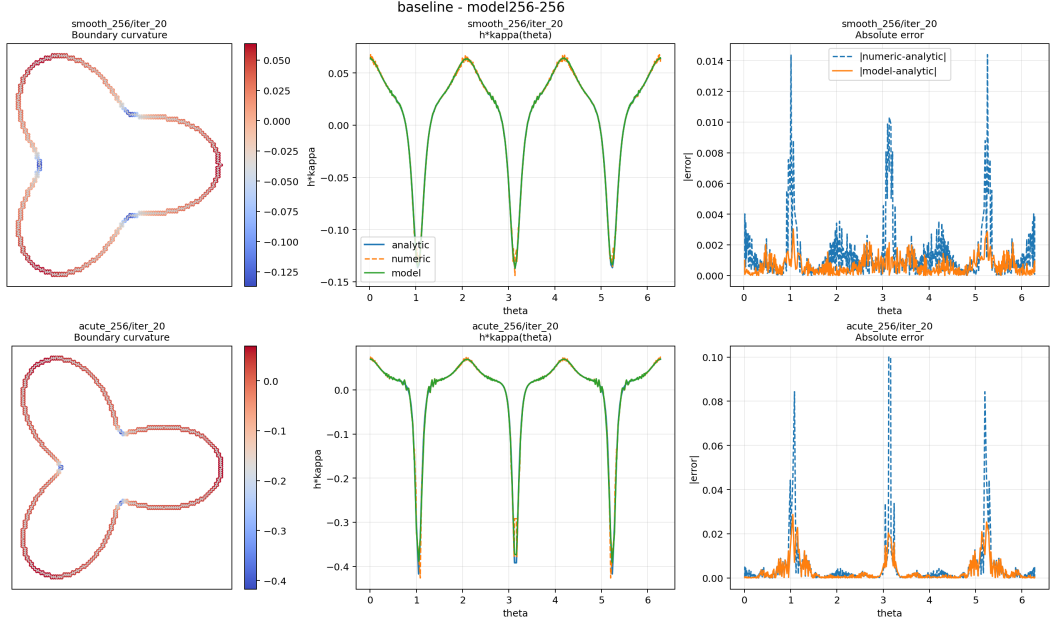


Figure 6: Flower-interface curvature prediction of the raw- ϕ model at resolution 256. The model captures the global curvature profile, while the dominant errors occur near high-curvature tips.

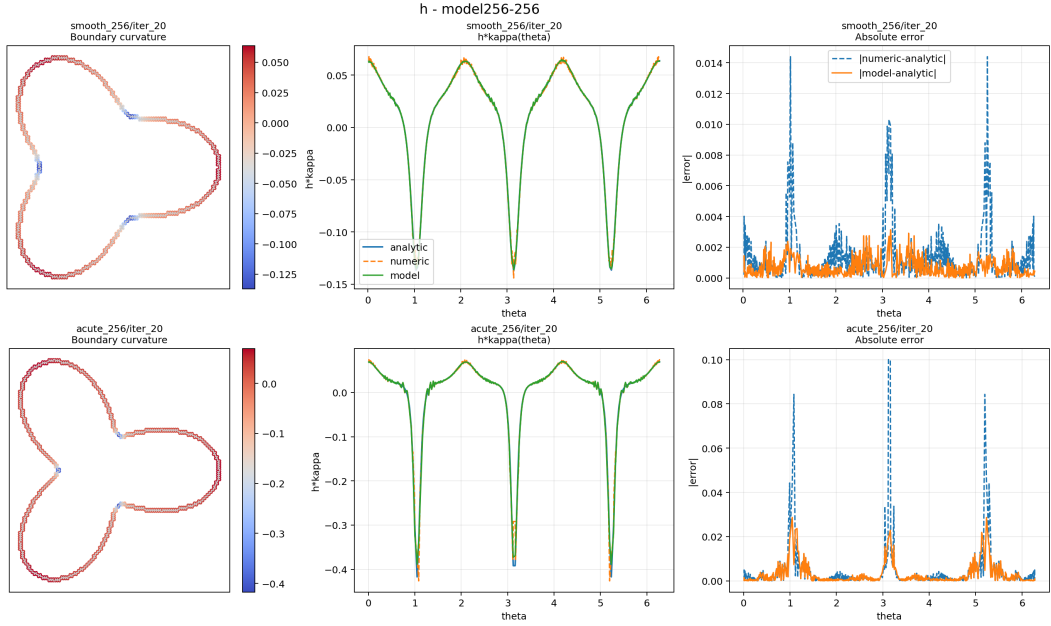


Figure 7: Flower-interface curvature prediction of the normalized- ϕ/h model at resolution 256. The overall behavior is close to the raw- ϕ baseline, with residual errors still concentrated near high-curvature tip regions.

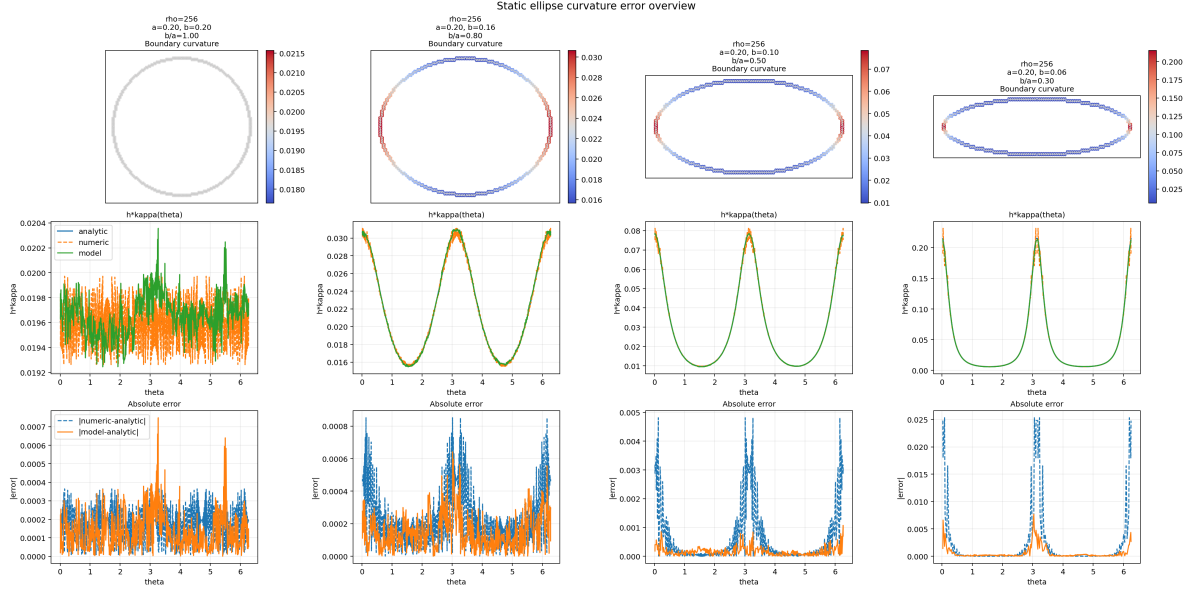


Figure 8: Static ellipse curvature prediction of the baseline stencil model at resolution 256. As the eccentricity increases, the curvature and the corresponding prediction errors become increasingly localized near the high-curvature endpoints.

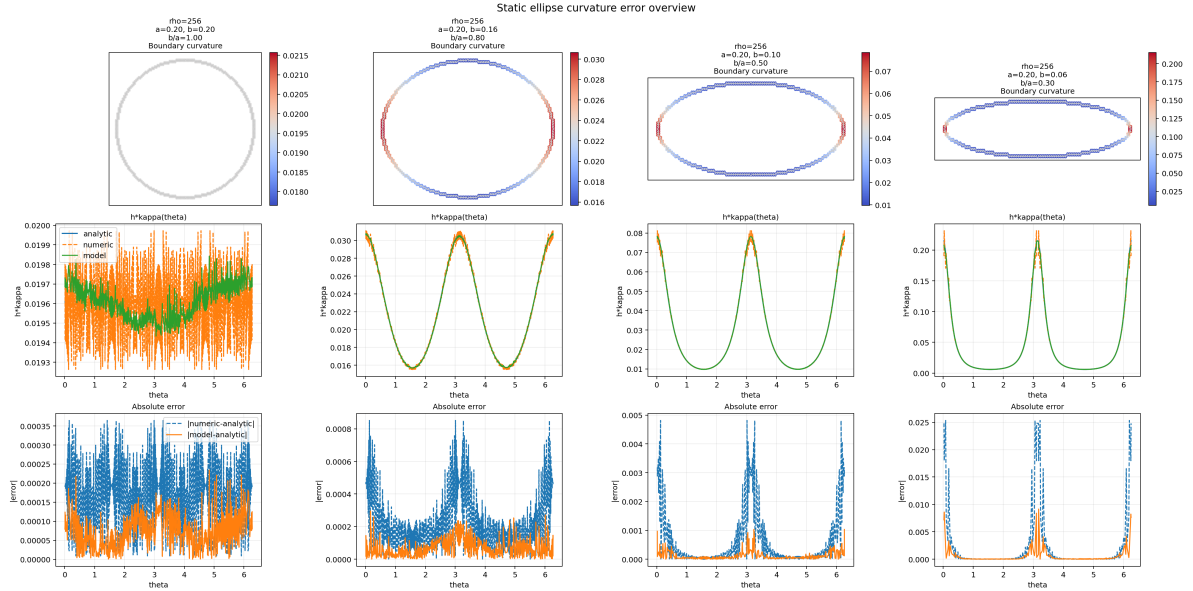


Figure 9: Static ellipse curvature prediction of the gradient-augmented model at resolution 256. The overall error is reduced compared with the baseline model, while the dominant residuals remain localized near the high-curvature endpoints.

approximation quality, but do not fully resolve the local difficulty of curvature prediction in rapidly varying regions.

The comparison indicates that the direct inclusion of gradient features does not necessarily improve curvature prediction near geometrically singular or near-singular regions. Although the augmented input provides additional first-order local information, the resulting model exhibits larger error peaks around the high-curvature tips. This behaviour is particularly evident for the acute flower interface, where the curvature varies rapidly along the boundary.

A possible explanation is that first-order gradient information improves local orientation awareness but does not directly constrain the second-order curvature structure. Consequently, the model may become more sensitive to local variations of the level-set field near sharp tips, leading to amplified prediction errors in precisely the regions where curvature estimation is most ill-conditioned.

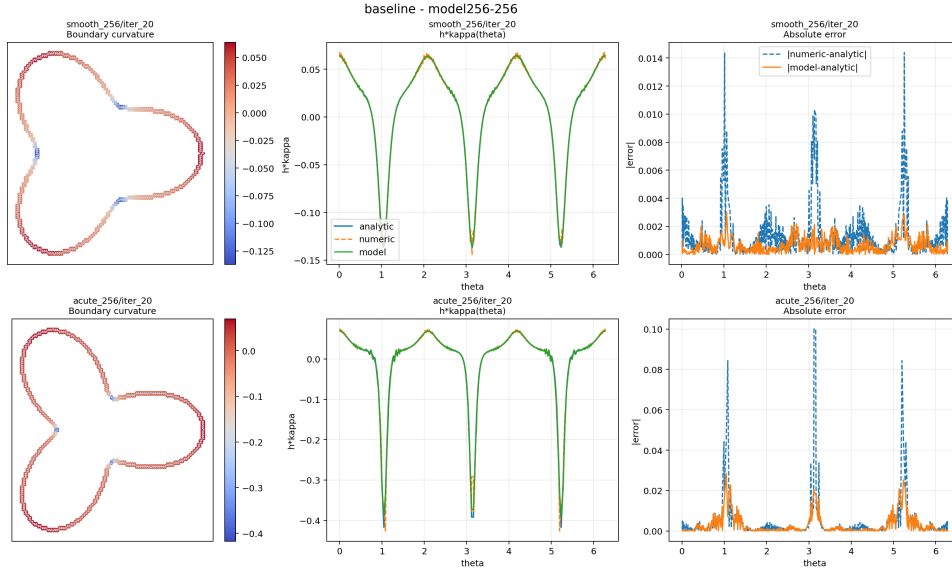


Figure 10: Prediction results of the baseline stencil model on smooth and acute flower-shaped interfaces at resolution 256. The model reproduces the global curvature profile, while the dominant discrepancies are localized near high-curvature tip regions.

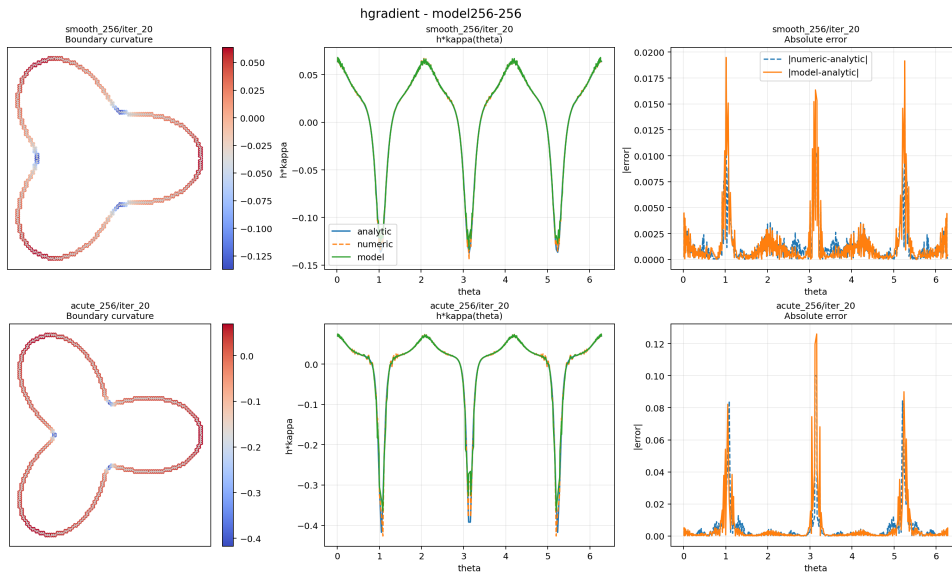


Figure 11: Prediction results of the gradient-augmented model on the same flower-shaped interfaces. Compared with the baseline model, the inclusion of gradient features leads to larger localized error peaks near the high-curvature tips, especially in the acute flower case.

4.1.4 Generalizability

Figures 12 and 13 compare cross-resolution generalization from $\rho_{\text{train}} = 128$ to $\rho_{\text{test}} = 256$ on static ellipses. In both cases, the predicted $h\kappa(\theta)$ remains close to the analytic profile across the ellipse family, indicating that the learned mapping generalizes across resolutions.

A clear difference is observed in the error distributions. The normalized- h model already yields accurate predictions for the more eccentric ellipses, but its error is relatively oscillatory in the circular and mildly eccentric cases. By contrast, the gradient-augmented model produces consistently lower and more stable model-to-analytic errors for $b/a = 1.0$ and $b/a = 0.8$, while maintaining similarly good performance for $b/a = 0.5$ and $b/a = 0.3$.

Therefore, both models exhibit cross-resolution generalizability, but the gradient-augmented model shows better overall robustness. In both models, the remaining residuals are still mainly concentrated near the high-curvature endpoints.

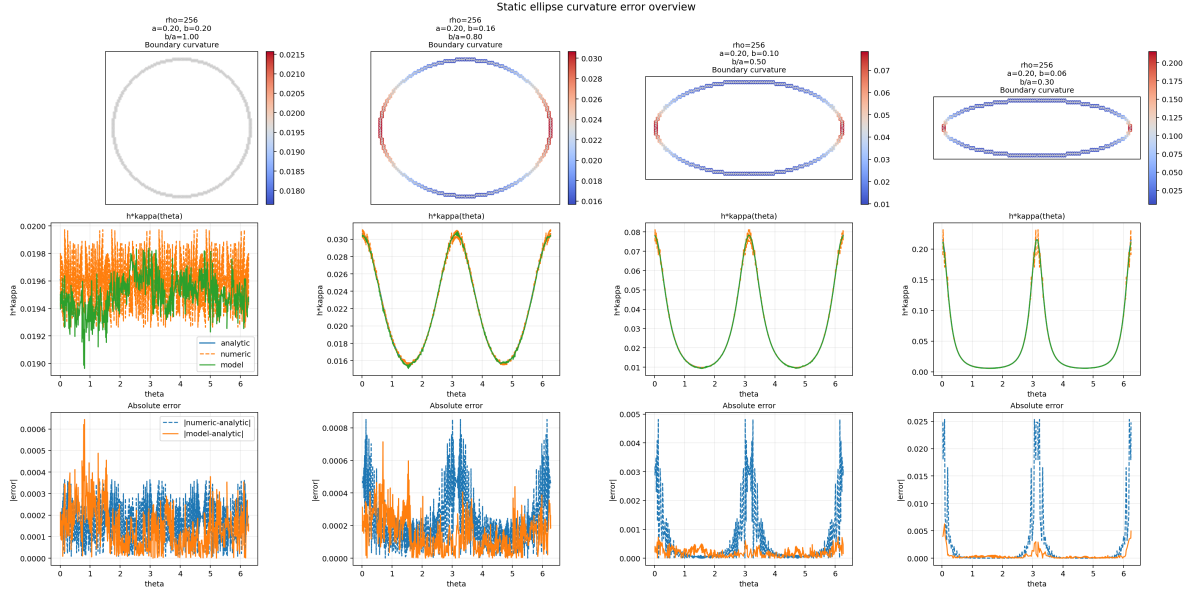


Figure 12: Cross-resolution ellipse evaluation of the normalized- h model, trained at $\rho = 128$ and tested at $\rho = 256$.

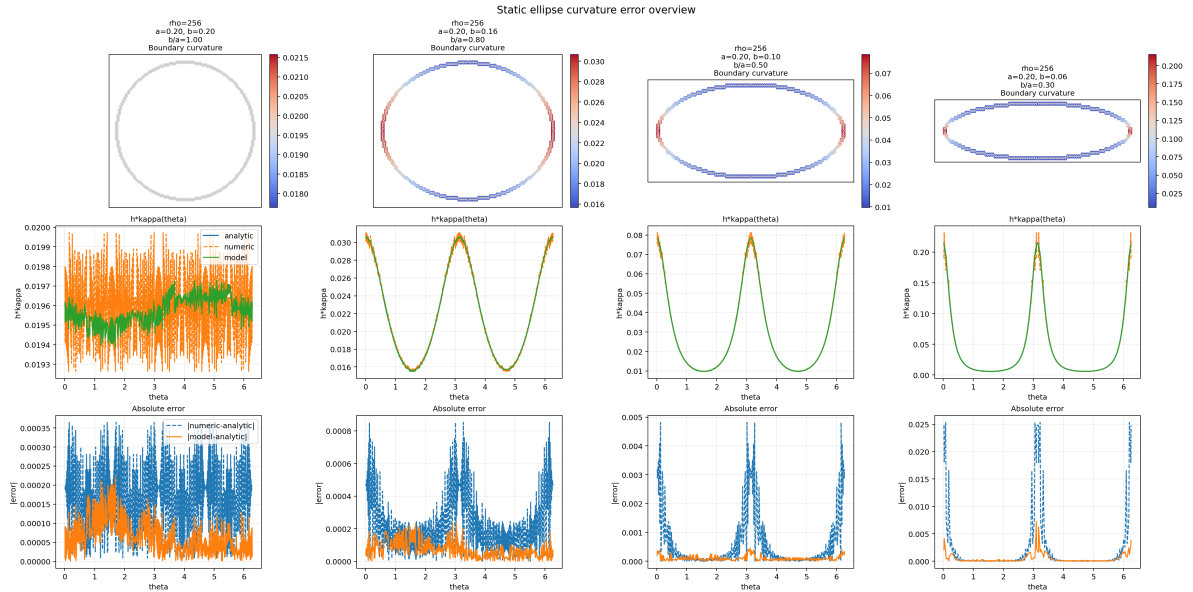


Figure 13: Cross-resolution ellipse evaluation of the gradient-augmented model, trained at $\rho = 128$ and tested at $\rho = 256$.

4.2 Summary

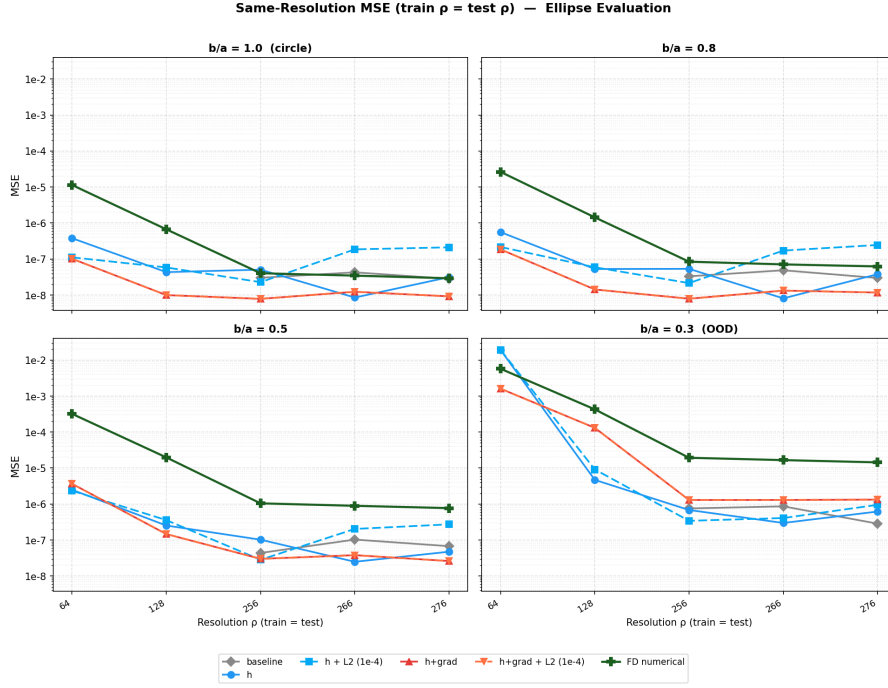


Figure 14: Same-resolution ellipse evaluation with $\rho_{\text{train}} = \rho_{\text{test}}$ for different aspect ratios b/a .

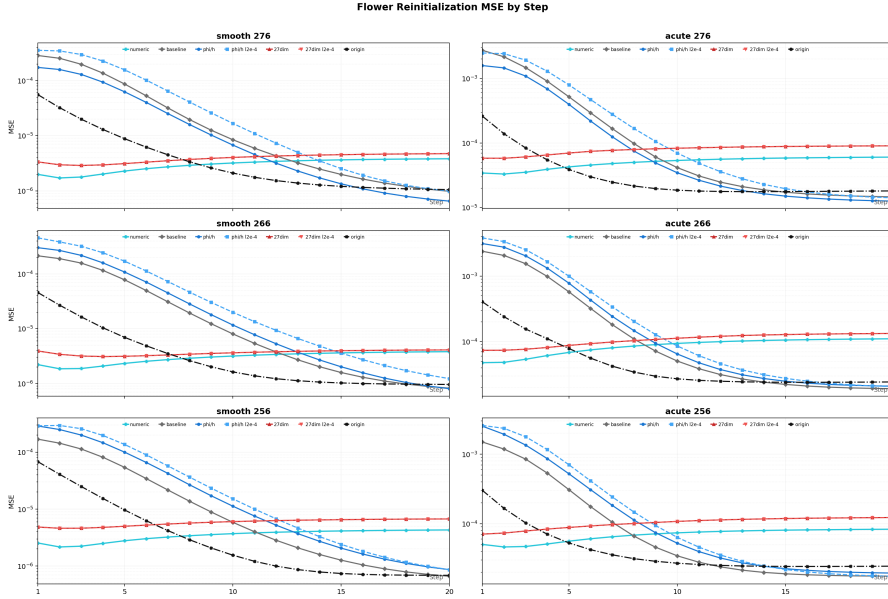


Figure 15: Reinitialization-step MSE on smooth and acute flower interfaces at resolutions 256, 266, and 276.

Summary of ellipse evaluation. Figure 14 evaluates same-resolution curvature prediction on ellipses with different aspect ratios b/a .

- For in-distribution ellipses, i.e. $b/a = 1.0, 0.8, 0.5$, the learned models achieve lower MSE than the finite-difference numerical baseline at moderate and high resolutions.
- As the ellipse becomes more eccentric, the prediction task becomes harder. This is reflected by the larger MSE in the $b/a = 0.3$ case, which is treated as an out-of-distribution geometry.

- Increasing the grid resolution generally reduces the error for all methods, but the improvement is less uniform in the OOD case. This indicates that resolution refinement improves curvature approximation, while strong geometric anisotropy remains a limiting factor.
- Gradient-based input features improve the overall error level in several ellipse cases, but the gain is not uniformly dominant across all aspect ratios and resolutions.

Summary of flower reinitialization evaluation. Figure 15 evaluates the MSE evolution along the reinitialization process for smooth and acute flower interfaces.

- The learned models generally show decreasing MSE as the reinitialization step increases, indicating that the prediction becomes more accurate as the level-set field approaches a more regular signed-distance structure.
- The acute flower cases consistently produce larger residual errors than the smooth flower cases. This confirms that high-curvature and rapidly varying interface regions dominate the difficulty of curvature prediction.
- The numerical reference error is comparatively stable or slowly increasing over reinitialization steps, while several learned models exhibit a clear error-decay trend.
- Across the three resolutions, the qualitative behavior is consistent: smooth geometries are easier, acute geometries are harder, and the main model limitation remains the robustness near sharp geometric features.

Overall, Figures 14 and 15 show that the learned models improve the global curvature approximation in both static ellipse and dynamic flower tests. However, the remaining errors are still mainly controlled by geometric difficulty, especially strong eccentricity and localized high-curvature regions.

5 Experiment 2

5.1 Setting

In this experiment, we introduce a scale factor $a > 0$ and use the effective grid spacing ah to simulate multiscale grid resolutions. The input stencil is normalized by ah , while the curvature target is scaled as $ah\kappa$. In addition, we use PCA to reduce the 27-dimensional augmented input to an 18-dimensional representation, so that the model can retain more geometric information while suppressing noisy or redundant components.

- the scale factor a is a constant, $a \in \{0.25, 0.5, 1.0\}$.

1. Normalized stencil model with ah . The first model scales the nine-point stencil by the effective grid spacing ah :

$$\mathbf{x}^{(m)} = \frac{\phi_9}{ah} \in \mathbb{R}^9, \quad y^{(m)} = ah\kappa \in \mathbb{R}. \quad (33)$$

Compared with the normalized stencil model in Experiment 1 Eqs 30, the additional factor a changes the effective grid scale and allows the same formulation to represent multiple resolutions.

2. 27-dimensional augmented model with ah . The second model augments the normalized stencil $\phi_9/(ah)$ with the unit-normal components at the nine stencil nodes:

$$\mathbf{x}^{(m)} = \left(\frac{\phi_9}{ah}, \mathbf{n}_9^x, \mathbf{n}_9^y \right) \in \mathbb{R}^{27}, \quad y^{(m)} = ah\kappa \in \mathbb{R}. \quad (34)$$

Here $\mathbf{n}_9^x$ and $\mathbf{n}_9^y$ are computed in the same way as in Experiment 1 Eqs 31. This 27-dimensional input combines the signed-distance stencil and local normal information, providing richer geometric features for curvature prediction.

3. PCA-reduced 18-dimensional model. The third model starts from the 27-dimensional augmented feature vector

$$\mathbf{z}^{(m)} = \left(\frac{\phi_9}{ah}, \mathbf{n}_9^x, \mathbf{n}_9^y \right) \in \mathbb{R}^{27}, \quad (35)$$

and applies PCA to project it onto an 18-dimensional feature space:

$$\mathbf{x}^{(m)} = \mathbf{P}_{18}^\top (\mathbf{z}^{(m)} - \boldsymbol{\mu}) \in \mathbb{R}^{18}, \quad y^{(m)} = ah\kappa \in \mathbb{R}. \quad (36)$$

Here $\boldsymbol{\mu} \in \mathbb{R}^{27}$ is the mean feature vector computed from the training set, and $\mathbf{P}_{18} \in \mathbb{R}^{27 \times 18}$ contains the first 18 principal components. This model keeps the main information from the 27-dimensional geometric features while reducing input noise and redundancy.

PCA is fitted on the training split after feature-wise standardization:

$$\begin{aligned} \mu_j &= \frac{1}{N_{\text{train}}} \sum_{m=1}^{N_{\text{train}}} z_j^{(m)}, & \hat{\sigma}_j^2 &= \frac{1}{N_{\text{train}} - 1} \sum_{m=1}^{N_{\text{train}}} (z_j^{(m)} - \mu_j)^2, \\ \sigma_j &= \begin{cases} \hat{\sigma}_j, & \hat{\sigma}_j > 0, \\ 1, & \hat{\sigma}_j = 0, \end{cases} & \tilde{\mathbf{z}}^{(m)} &= D_\sigma^{-1} (\mathbf{z}^{(m)} - \boldsymbol{\mu}), & D_\sigma &= \text{diag}(\sigma_1, \dots, \sigma_{27}), \\ \tilde{\mathbf{Z}}_{\text{train}} &= \begin{bmatrix} (\tilde{\mathbf{z}}^{(1)})^\top \\ \vdots \\ (\tilde{\mathbf{z}}^{(N_{\text{train}})})^\top \end{bmatrix}, & C &= \frac{1}{N_{\text{train}} - 1} \tilde{\mathbf{Z}}_{\text{train}}^\top \tilde{\mathbf{Z}}_{\text{train}}, \\ C\mathbf{q}_r &= \lambda_r \mathbf{q}_r, & \lambda_1 &\geq \lambda_2 \geq \dots \geq \lambda_{27} \geq 0. \end{aligned}$$

Let

$$\mathbf{Q}_{18} = [\mathbf{q}_1 \quad \mathbf{q}_2 \quad \dots \quad \mathbf{q}_{18}] \in \mathbb{R}^{27 \times 18}.$$

The PCA-reduced input and target are then

$$\mathbf{x}^{(m)} = \mathbf{Q}_{18}^\top \tilde{\mathbf{z}}^{(m)} \in \mathbb{R}^{18}, \quad y^{(m)} = ah\kappa^{(m)} \in \mathbb{R}, \quad \frac{\sum_{r=1}^{18} \lambda_r}{\sum_{r=1}^{27} \lambda_r} \geq 0.95.$$

Here $\boldsymbol{\mu}$, D_σ , and $\mathbf{Q}_{18}$ are computed once from the training split and reused unchanged for validation and evaluation.

5.2 Analysis

5.2.1 Grid generalization

- The raw ϕ representation is not grid-invariant and shows poor cross-resolution robustness.
- Scaling the level-set values by h improves generalization, confirming that curvature learning requires a dimensionless local representation.
- Incorporating local normal information (n_x, n_y) further stabilizes the prediction, especially under resolution transfer and complex flower geometries.
- The feature set $(\phi/h, n_x, n_y)$ gives the most reliable performance across grid resolutions, reinitialization steps, and geometry types.
- The PCA-reduced model preserves the accuracy of the full 27D representation, indicating that the learned geometric features contain significant redundancy.

Figure 16 reports the test-split MSE of different feature representations under cross-resolution evaluation. The baseline finite-difference curvature shows the expected resolution-dependent behavior, with the error decreasing as the grid is refined. However, its accuracy remains limited compared with the best learning-based models.

Figure 17 shows the flower-interface results for models trained at $\rho = 128$. The 9D models based only on ϕ/h show clear sensitivity to both reinitialization step and test resolution, especially for the acute flower case. This indicates that the local level-set stencil alone is not sufficient for robust curvature prediction on complex geometries.

By contrast, the models using $(\phi/h, n_x, n_y)$ give consistently lower and more stable errors across resolutions and reinitialization steps. The improvement is particularly clear at higher test resolutions, where the 27D feature representation remains accurate while the 9D models fluctuate more strongly.

Test-split MSE vs resolution — three variants

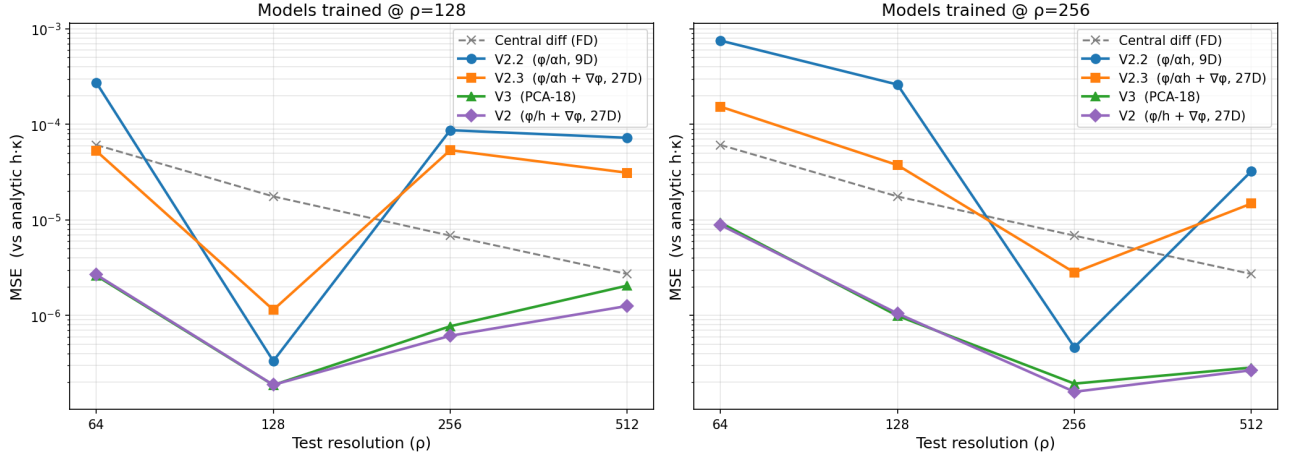


Figure 16: Grid generalization results for models trained at $\rho = 128$ and $\rho = 256$, and tested on resolutions $\rho = 64, 128, 256, 512$.

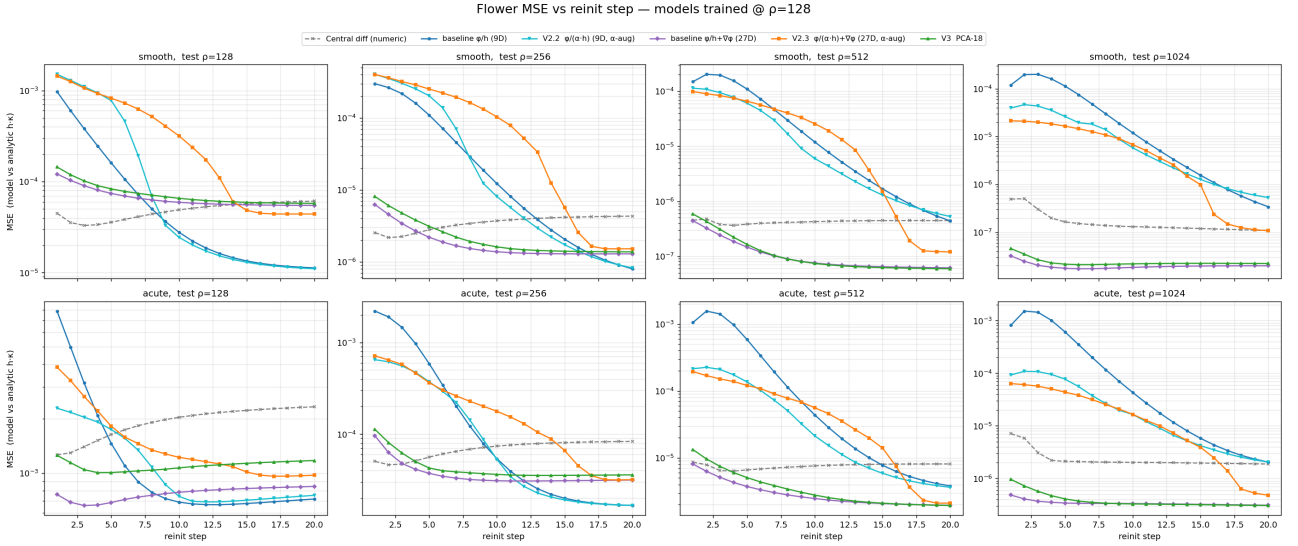


Figure 17: Flower-interface MSE over reinitialization steps for models trained at $\rho = 128$.

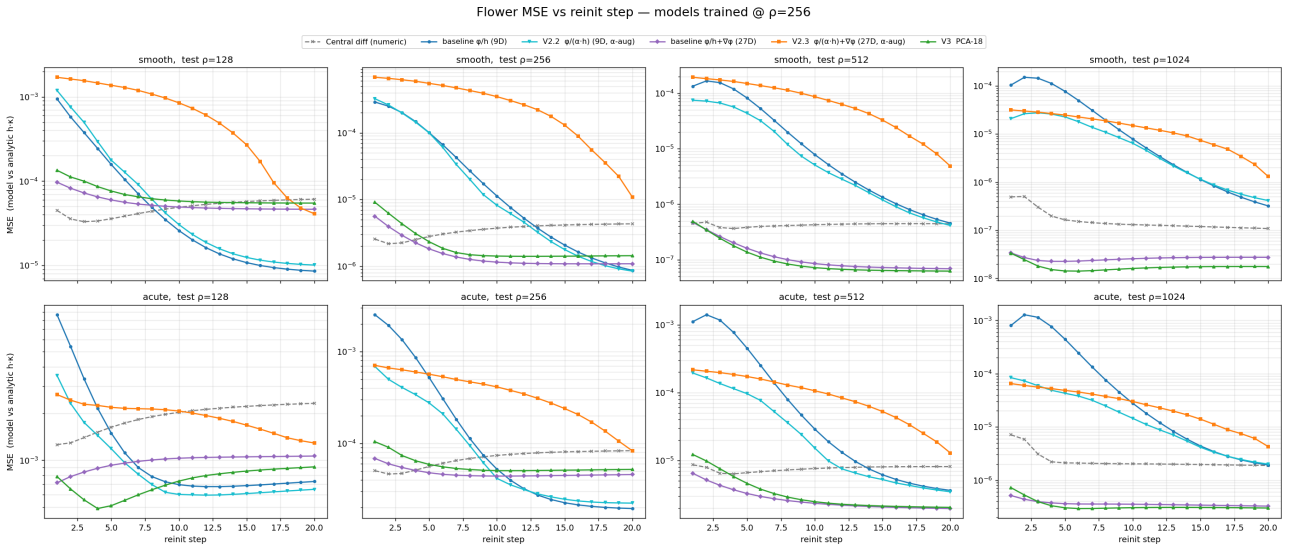


Figure 18: Flower-interface MSE over reinitialization steps for models trained at $\rho = 256$.

Figure 18 shows a similar trend for models trained at $\rho = 256$. The geometry-aware models remain stable for both smooth and acute flowers, whereas the 9D models are more dependent on the reinitialization process. The PCA-18 model closely follows the full 27D model, suggesting that the curvature-relevant information in $(\phi/h, n_x, n_y)$ can be compressed without significant loss of accuracy.

Overall, the flower tests confirm that reliable generalization requires dimensionless and geometry-aware inputs. In particular, $(\phi/h, n_x, n_y)$ is more robust than using ϕ/h alone, and PCA provides an efficient reduced representation of the same geometric information.

5.2.2 27D-PCA

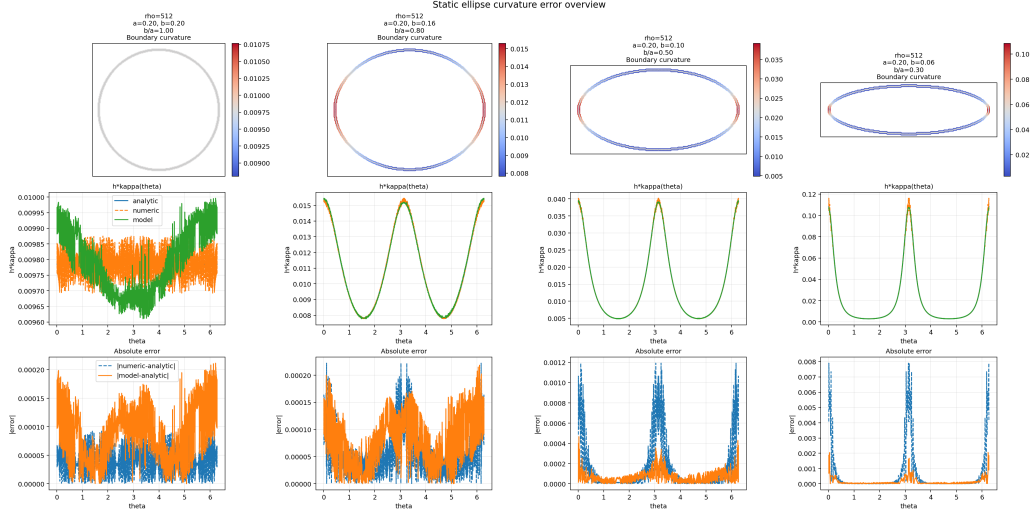


Figure 19: Static ellipse curvature prediction using the full 27D feature representation. The model is trained at $\rho = 256$ and tested at $\rho = 512$.

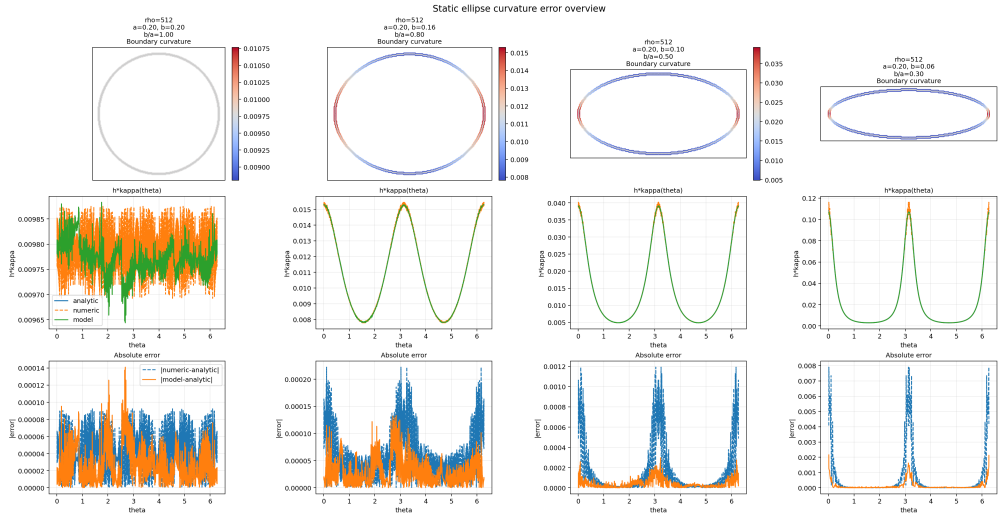


Figure 20: Static ellipse curvature prediction using the PCA-reduced feature representation. The model is trained at $\rho = 256$ and tested at $\rho = 512$.

Figures 19 and 20 compare the full 27D representation with its PCA-reduced version on static ellipses. For ellipses with clear curvature variation, both models follow the analytic curvature profile closely. The dominant errors are concentrated near the ellipse tips, where the curvature changes most rapidly. This is also where the finite-difference baseline has its largest error, indicating that these regions are intrinsically more difficult for curvature approximation.

The PCA model preserves the main behavior of the full 27D model on these ellipse tests. However, the nearly circular case shows that PCA can introduce a small systematic deviation when the target curvature is almost constant. Therefore, PCA should be interpreted as an efficient compression of the 27D geometry-aware features, rather than a strictly lossless replacement.

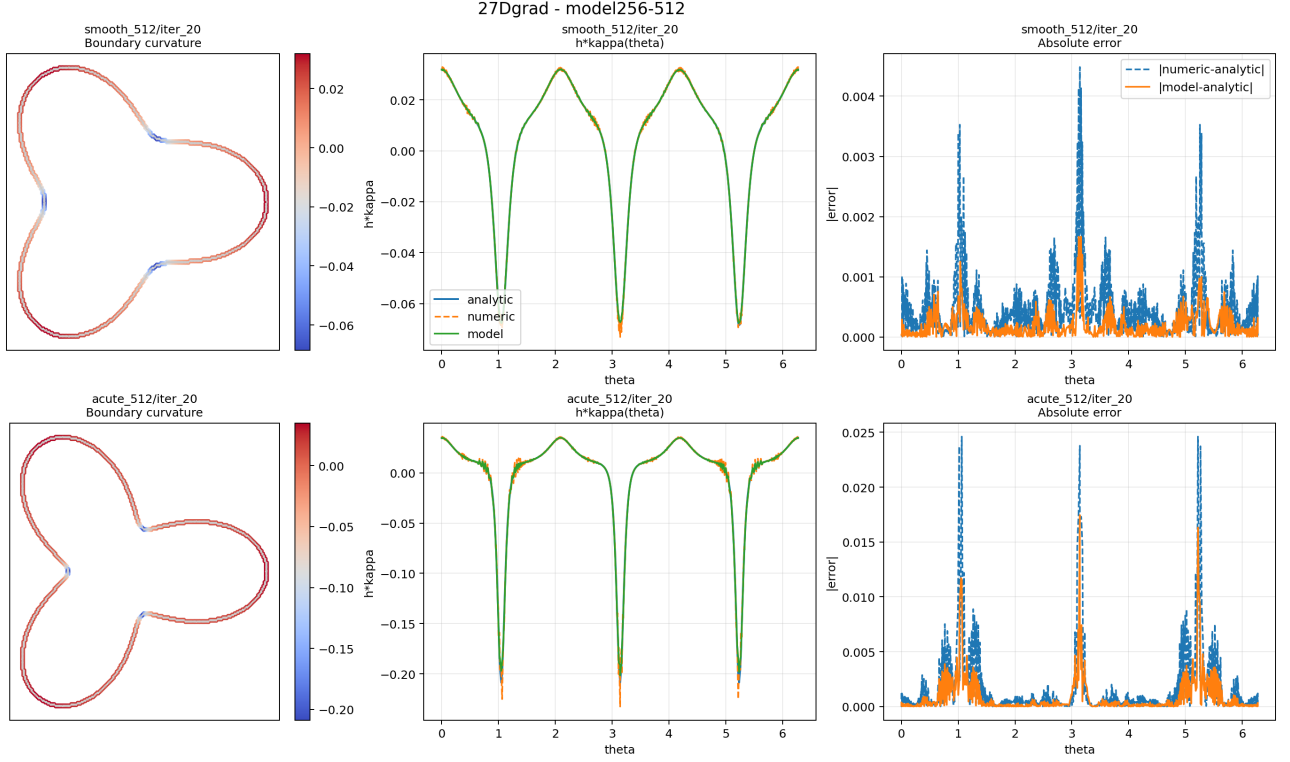


Figure 21: Flower-interface curvature prediction using the full 27D feature representation. The model is trained at $\rho = 256$ and tested at $\rho = 512$.

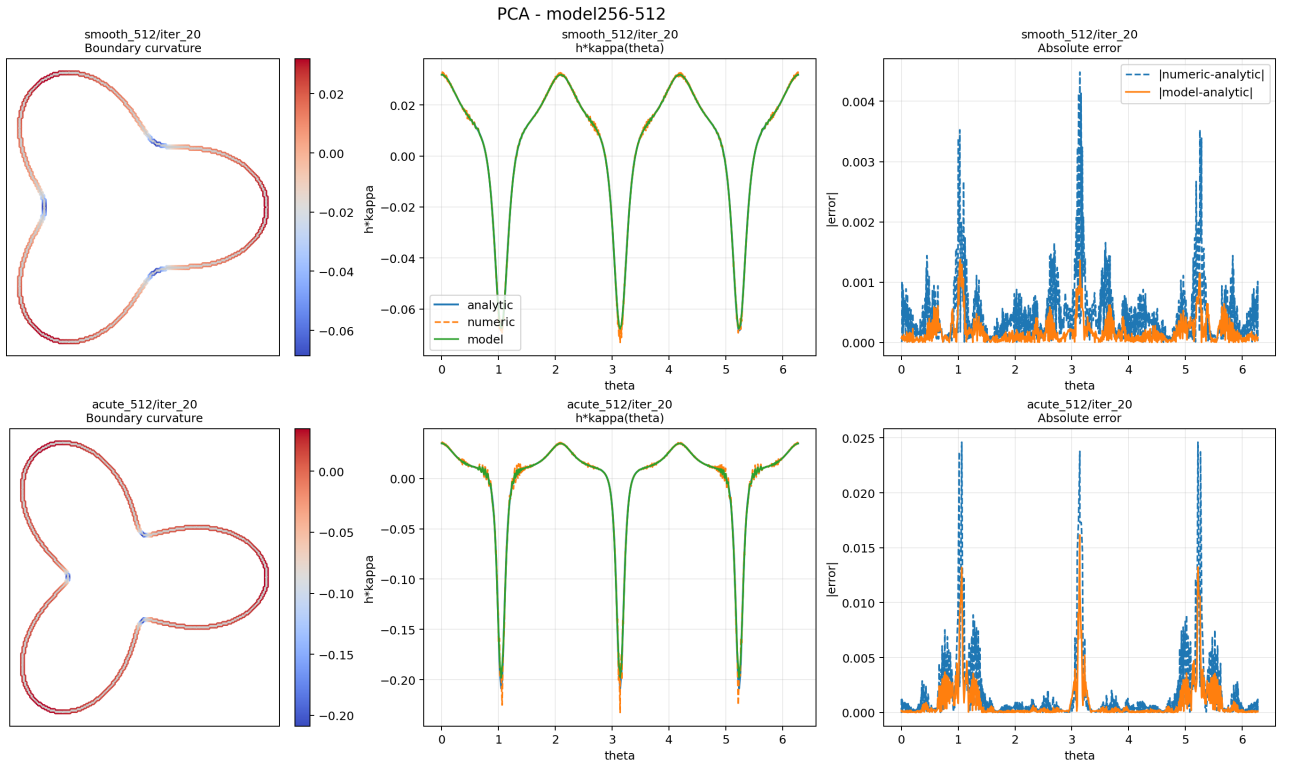


Figure 22: Flower-interface curvature prediction using the PCA-reduced feature representation. The model is trained at $\rho = 256$ and tested at $\rho = 512$.

Figures 21 and 22 evaluate the same comparison on flower interfaces. Compared with the ellipse cases, the flower geometries contain stronger concave regions and sharper curvature variations. Both the full 27D and PCA models recover the main analytic curvature profile, while the largest errors remain localized around the negative curvature peaks. These peak locations are consistent with the regions where the numerical baseline also deteriorates.

The comparison indicates that PCA does not substantially change the failure pattern of the 27D model. It preserves the main curvature structure and the locations of the difficult regions, but it may slightly smooth or bias local details. Overall, PCA is effective as a reduced representation of $(\phi/h, n_x, n_y)$: it keeps most of the predictive information of the full 27D input, while its limitations appear mainly in nearly constant-curvature or highly localized high-curvature regions.

5.2.3 flower test with all models

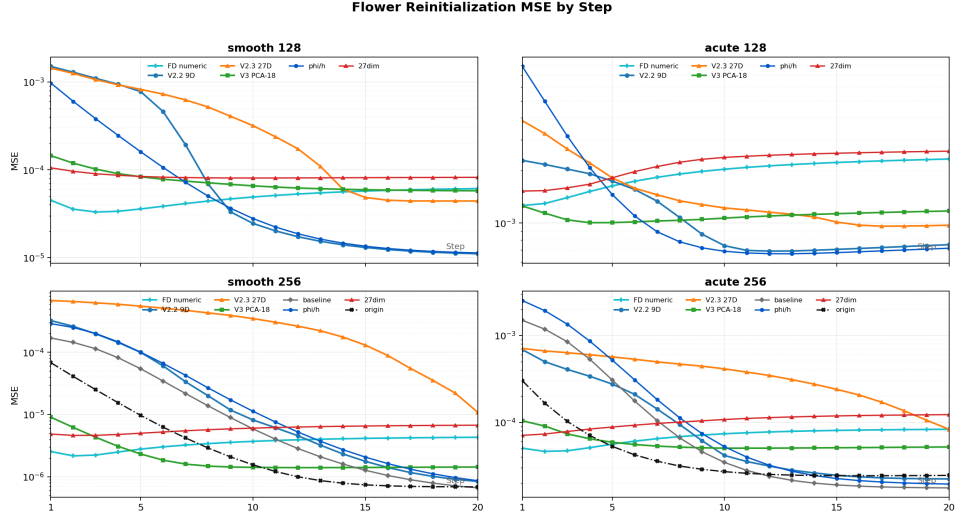


Figure 23: flower test